

Agronomic Knowledge Retrieval

Problem Statements for a Retrieval-Augmented Generation (RAG) Challenge — from Nigeria to Africa to a Global Context

PROBLEM STATEMENT 1

Localized Agronomic Knowledge Retrieval in Nigeria

Scope	National — Nigeria
Input	Natural language farmer query (crop symptoms, soil conditions)
Output	Top-5 most relevant documents from a curated agricultural knowledge base
Baseline	TF-IDF
Target Techniques	BM25, dense embeddings, hybrid search, reranking
Evaluation Metric	nDCG@5 against expert-labelled relevance judgments

Context

Smallholder farmers in Nigeria form the backbone of the country's agricultural production, contributing significantly to food security and rural livelihoods. However, these farmers often lack timely access to accurate, localized agronomic advice. Challenges such as crop diseases, pest infestations, soil degradation, and unpredictable climate patterns frequently go unaddressed due to limited extension services, language barriers, and poor information dissemination channels. While agricultural knowledge exists in extension documents, research publications, and government guidelines, retrieving the most relevant information for a farmer's specific query remains a major bottleneck.

Task Definition

This problem focuses on designing an optimized document retrieval system at the core of a Retrieval-Augmented Generation (RAG) pipeline. Given a natural language query from a Nigerian smallholder farmer — such as symptoms observed in crops or soil conditions — the system must return the top-5 most relevant documents from a curated agricultural knowledge base. The retrieval system should account for local context, including region-specific crops (e.g., cassava, maize, rice), common pests and diseases, and climate conditions unique to Nigeria's agro-ecological zones.

Technical Challenge

The baseline TF-IDF approach often fails to capture semantic meaning, synonymy, and contextual nuances present in farmer queries. Therefore, this problem challenges participants to implement more advanced retrieval techniques such as BM25, dense vector embeddings, hybrid search, or reranking models to improve relevance ranking. Performance will be evaluated using nDCG@5 against expert-labelled relevance judgments.

Impact

Solving this problem will enable scalable, intelligent access to agronomic knowledge, empowering Nigerian farmers with precise, context-aware recommendations and reducing dependency on limited human extension services.

PROBLEM STATEMENT 2

Scalable Agronomic Retrieval for Africa and Global Food Systems

Scope	Continental / Global
Input	Farmer query across diverse linguistic, cultural, and ecological contexts
Output	Ranked relevant documents from a heterogeneous global knowledge base
Baseline	TF-IDF (insufficient for cross-lingual, domain-specific nuance)
Target Techniques	Sentence-transformer embeddings, hybrid retrieval, learning-to-rank rerankers
Evaluation Metric	nDCG@5

Context

Across Africa and the broader global agricultural landscape, smallholder farmers face increasingly complex challenges driven by climate change, soil nutrient depletion, emerging pests and diseases, and limited access to agronomic expertise. Despite the availability of extensive agricultural knowledge repositories — from international research bodies, NGOs, and government agencies — there remains a critical gap in delivering the right information to the right farmer at the right time. This gap is particularly pronounced in low-resource settings where digital infrastructure is limited and extension services are overstretched.

Task Definition

This problem addresses the need for a scalable, intelligent retrieval system that can serve as the backbone of a global Retrieval-Augmented Generation (RAG) framework for agronomic advice. Given a farmer's query — potentially from diverse linguistic, cultural, and ecological contexts — the system must identify and rank the most relevant documents from a heterogeneous knowledge base covering crops, soil health, pest management, fertilization strategies, and climate adaptation practices.

Technical Challenge

Unlike region-specific systems, this challenge requires models that generalize across different geographies, crops, and farming practices while still maintaining high relevance. Traditional keyword-based methods like TF-IDF are insufficient for capturing cross-lingual semantics, domain-specific terminology, and nuanced agricultural conditions. Participants are encouraged to explore advanced retrieval strategies such as dense embeddings (e.g., sentence transformers), hybrid retrieval pipelines, and learning-to-rank rerankers to improve performance.

Impact

Evaluation will be conducted using nDCG@5, emphasizing the importance of ranking highly relevant documents in the top results. A successful solution will contribute to building robust, scalable AI systems that

democratize access to agricultural knowledge, improve decision-making for farmers, and ultimately strengthen food security across Africa and the world.